AI-Integrated Multi-Modal Academic Evaluation, Performance Intelligence, and Automated Viva Assessment System

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Project Proposal Report

L.S.B Hemarathne – IT22134776

B.Sc. (Hons) Degree in Information Technology Specialized in Software Engineering

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
L.S.B Hemarathne	**IT22134776**	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Ms. Sanjeevi Chandrasiri	15/03/2026	

ABSTRACT

The rapid growth of digital learning platforms and automated assessment systems has created large volumes of educational data. However, most current assessment systems primarily focus on grading student answers and providing final scores, offering limited insight into the deeper learning challenges faced by students. In many educational environments, instructors struggle to identify which exam questions are commonly misunderstood, which concepts students find difficult, and which cognitive skills require further development. Although research in Learning Analytics and Educational Data Mining has explored student performance prediction, there remains a significant gap in analyzing question-level misunderstandings and predicting future exam patterns based on historical assessment data.

To address this gap, this research proposes a Predictive Learning Analytics and Intelligent Question Analysis module that provides deep insights into student learning behavior. The system analyzes historical exam questions, student performance data, and concept clusters to identify frequently misunderstood questions, detect weak knowledge areas, and diagnose conceptual gaps. In addition, the proposed system applies pattern analysis and machine learning techniques to study past examination trends and predict potential future exam topics and question structures.

The solution will utilize technologies such as Natural Language Processing (NLP), machine learning models, and pattern mining techniques including Topic Modeling and Clustering (machine learning) to analyze educational data effectively.

The expected outcome is an intelligent analytics system capable of providing actionable insights for both students and educators. This system can support personalized learning, improve exam preparation strategies, and assist educators in refining teaching methods. From a social perspective, the proposed approach can enhance educational quality and learning outcomes, while commercially it has potential applications in modern Learning Management Systems and AI-powered educational platforms.

Keywords: Predictive Learning Analytics, Exam Pattern Prediction, Educational Data Mining, Question-Level Analytics, AI in Education.

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LIST OF ABBREVIATIONS

Table 1 Abbreviations Table

Abbreviation	**Full Meaning**
AI	Artificial Intelligence
AIED	Artificial Intelligence in Education
BERT	Bidirectional Encoder Representations from Transformers
DSR	Design Science Research
EDM	Educational Data Mining
FR	Functional Requirement
LA	Learning Analytics
LDA	Latent Dirichlet Allocation
LKR	Sri Lankan Rupees
LMS	Learning Management System
ML	Machine Learning
NLP	Natural Language Processing
PLA	Predictive Learning Analytics
RoBERTa	Robustly Optimized BERT Approach
SaaS	Software as a Service
SDG	Sustainable Development Goal

1. INTRODUCTION

1.1 Background and Context

The rapid integration of digital technologies in education has led to the widespread adoption of online learning platforms and Learning Management Systems (LMS). These systems generate large volumes of student assessment data through quizzes, assignments, and examinations. However, most educational platforms primarily focus on grading and storing student performance data, offering limited insights into deeper learning patterns. As a result, educators often find it difficult to identify which exam questions are commonly misunderstood, which concepts students struggle with most, and what cognitive skills require improvement [1].

Students are also affected by this limitation. Many learners prepare for examinations without clear insight into their conceptual weaknesses or the types of questions that frequently appear in exams. Consequently, students tend to rely on memorization rather than strategic learning based on data-driven insights. Studies show that the use of learning analytics technologies can significantly improve student performance by identifying learning gaps early and enabling targeted interventions [2]. However, many current systems still lack advanced question-level analysis and predictive capabilities.

Existing solutions within the fields of Learning Analytics and Educational Data Mining mainly focus on predicting student grades or identifying students at risk of academic failure. While these approaches provide useful insights, they rarely analyze the underlying exam questions themselves or identify patterns in how lecturers structure examinations. Additionally, most current systems do not provide predictive insights regarding future exam topics or question structures based on historical exam data [3].

To address these limitations, this research proposes a Predictive Learning Analytics and Intelligent Question Analysis module that performs deep question-level analytics. The system aims to identify frequently misunderstood questions, detect weak concepts among students, and analyse patterns in historical exam papers. Using techniques such as Natural Language Processing and machine learning approaches like Topic Modeling and Clustering, the proposed system will analyse exam content and student performance data to generate insights and predict likely future exam topics.

The proposed solution aligns with the goals of improving educational quality and supporting data-driven learning strategies. It also contributes to the objectives of the United Nations Sustainable Development Goals, particularly Sustainable Development Goal 4, which emphasizes the importance of inclusive and high-quality education supported by innovative technologies [4].

From a research perspective, this study contributes to the growing field of Artificial Intelligence in Education (AIED) by integrating predictive learning analytics with question-level exam analysis. The research aligns with academic clusters such as Artificial Intelligence, Data Science, and Educational Technology, aiming to develop intelligent systems that enhance both teaching effectiveness and student learning outcomes.

1.2 Literature Review

Recent developments in educational technology have significantly increased research interest in applying artificial intelligence and data analytics to improve assessment and learning processes. The fields of Learning Analytics and Educational Data Mining focus on extracting meaningful insights from large volumes of educational data such as student performance records, assessments, and learning interactions. Several studies have explored how machine learning and data mining techniques can be used to analyze educational datasets to support decision-making in academic environments.

One major area of research involves predicting student performance using machine learning models. Various techniques such as Random Forest, Neural Networks, and Naïve Bayes have been applied to analyze student learning outcomes and predict academic success. These models typically use features such as attendance, coursework scores, and interaction data to forecast student performance. However, many studies focus mainly on predicting final grades rather than analyzing exam questions themselves or identifying conceptual learning gaps [5].

Another research direction focuses on the application of Natural Language Processing (NLP) techniques in education. NLP allows systems to process and analyze textual educational data such as exam questions, student answers, and learning materials. Recent studies demonstrate that NLP can support applications such as automated question generation, intelligent tutoring systems, and automated assessment tools. These technologies help instructors generate quizzes, analyze written responses, and provide feedback to students more efficiently [6]. [1]

Several research works have also explored automated assessment systems. These systems utilize AI models to evaluate student answers, particularly for open-ended questions. Automated assessment systems reduce the workload of educators and allow faster feedback for students. A systematic review of automated assessment technologies identified numerous AI-based models that can analyze textual student responses and evaluate conceptual understanding [7].

In addition to assessment automation, some studies use deep learning and sentiment analysis to analyze student feedback and learning experiences. By applying neural networks and text analytics techniques, researchers have attempted to extract useful insights from student comments and course evaluations. These insights can assist institutions in improving teaching strategies and course structures [8].

Recent studies have also introduced machine learning frameworks for analyzing learning behavior and educational interaction data. For example, process mining and clickstream analysis have been used to understand how students interact with digital learning platforms. These methods help researchers identify patterns in student learning behavior and detect potential difficulties during the learning process [9].

In terms of educational tools and products, several AI-based platforms have emerged that support automated question generation and learning assistance. Platforms such as Quillionz, PrepAI, and Lumos Learning use natural language processing to generate questions, extract key topics, and assist students in studying course materials. These systems demonstrate the

potential of AI in education but typically focus on limited functionalities such as question generation or content summarization rather than comprehensive exam analytics [10].

Despite these advancements, several limitations remain in current systems. Most existing research focuses on predicting student performance rather than analyzing exam questions themselves. Additionally, many educational analytics systems treat student data as aggregated performance metrics and do not identify which specific questions or concepts are frequently misunderstood by students. Furthermore, existing AI-based educational platforms often perform isolated tasks such as question generation or grading but do not provide integrated analytics that combines exam pattern analysis, concept clustering, and predictive insights regarding future exam topics [6].

Therefore, there is a clear research gap for systems that integrate question-level analytics with predictive learning analytics. A unified intelligent framework capable of analyzing historical exam papers, detecting commonly misunderstood questions, identifying weak concepts, and predicting future exam patterns could significantly improve both teaching effectiveness and student learning strategies. The proposed research aims to address these limitations by combining Natural Language Processing, machine learning clustering methods, and predictive analytics to build an intelligent exam analysis system.

1.3 Research Gap

Although developments in Predictive Learning Analytics and Educational Data Mining have significantly advanced educational technologies, a substantial research gap remains in extracting deep insights from historical examination data and predicting future examination trends. Current research has extensively focused on student performance prediction, automated grading, and learning behavior analysis. However, limited work has been conducted on analyzing the content of exam questions themselves while simultaneously incorporating historical student performance data. Specifically, there is a lack of an integrated framework that simultaneously (1) applies Natural Language Processing (NLP), topic modeling, and clustering techniques to historical exam question texts to identify conceptual clusters, (2) overlays student response and performance data to detect frequently misunderstood questions and diagnose conceptual gaps, and (3) utilizes predictive analytics to forecast potential future exam topics, question structures, and difficulty levels.

Existing research approaches remain insufficient for several key reasons. First, the majority of Predictive Learning Analytics studies are primarily student-outcome-centric. Systematic reviews indicate that most machine learning models focus on predicting final grades, identifying at-risk students, or forecasting academic performance using learning management system interaction logs and demographic data. While these approaches provide valuable predictive insights, they rarely analyze exam question structures or identify the specific questions that contribute to student learning difficulties [11].

Second, several studies have explored the use of automated classification techniques to categorize exam questions according to cognitive levels such as Bloom's Taxonomy. Machine learning models including Support Vector Machines, deep neural networks, and transformer-based language models have demonstrated high accuracy in classifying question difficulty and cognitive complexity. However, these systems typically analyze question text independently

and do not correlate question characteristics with historical student performance data. As a result, they cannot identify which questions consistently lead to misunderstanding or poor performance among students [12].

Third, topic modeling and clustering techniques have been applied to educational datasets to discover latent learning patterns and thematic structures within online learning environments. Nevertheless, these techniques are rarely applied directly to structured exam question banks combined with student performance datasets. Additionally, existing studies have not explored how historical exam patterns can be used to predict future examination trends or identify recurring question themes within academic assessments [13].

Furthermore, some recent research has attempted to improve intelligent assessment systems using machine learning and natural language processing. For example, the AsseslyAI framework uses historical laboratory question datasets and machine learning models to generate new questions and evaluate student responses. While this approach demonstrates the effectiveness of AI-driven assessment tools, it primarily focuses on skill-based laboratory assessments rather than comprehensive analysis of exam questions and conceptual misunderstandings [14].

Similarly, Transformer-based knowledge tracing models have been proposed to analyze sequences of student-question interactions and predict student performance on future exercises. These approaches are effective in modeling student learning progression but do not analyze the semantic content of exam questions or identify patterns in examination design [15].

Therefore, a clear research gap exists in developing an integrated intelligent framework that combines question-level textual analysis, student performance data, and predictive modeling for examination analytics. Existing systems typically address isolated aspects such as performance prediction, question classification, or automated assessment, but they do not provide a unified framework capable of identifying misunderstood concepts, clustering exam topics, and predicting potential future examination patterns.

To address this limitation, the proposed research introduces an AI-driven intelligent examination analytics framework that integrates Natural Language Processing, topic modeling, clustering techniques, and predictive learning analytics. The framework will analyze historical exam papers and student performance data to detect conceptual gaps, identify frequently misunderstood questions, cluster related exam topics, and forecast likely future exam themes. By combining these analytical components into a unified system, the proposed approach aims to provide deeper insights into examination structures while supporting both instructors and students with data-driven learning strategies.

This research contributes to the evolving domain of Artificial Intelligence in Education by introducing a comprehensive, question-centric predictive analytics framework capable of transforming historical educational data into actionable insights for curriculum improvement and personalized learning.

Table 2 Research gap comparision

Study / System	Research Focus	Methods Used	Limitations	Identified Research Gap
Romero & Ventura [16]	Educational Data Mining for student performance prediction	Machine learning models on LMS data	Focuses mainly on predicting final grades rather than analysing exam questions	Lack of question-level analytics and concept gap identification
Mohammed & Omar [17]	Automatic classification of exam questions based on Bloom's Taxonomy	NLP techniques such as TF-IDF and Word2Vec	Classifies cognitive levels of questions but ignores student performance analysis	No integration of question classification with learning analytics
Sucipto et al. [18]	Question classification using Bloom's Taxonomy	Data mining and machine learning approaches	Focuses on taxonomy classification only	Does not analyze historical exam trends or question difficulty patterns
Kumar et al. [19]	Analysis of examination data using Bloom's Taxonomy	Data mining and statistical analysis	Exam analysis limited to taxonomy distribution	Lack of predictive analytics for future exam questions
Prasad [20]	Identification of Bloom's taxonomy levels in question papers	NLP tokenization and text processing	Identifies question levels but does not analyse learning gaps	No connection with student performance data
Deokar et al. [21]	Automatic exam question generation using Bloom's taxonomy	Algorithm-based question generation	Focuses on generating exam questions	Does not analyse historical exam datasets or student errors
Alghamdi & Alghamdi [22]	Learning analytics for improving higher education outcomes	Educational data mining and analytics	Focuses on overall learning behaviour	Lack of question-level analytics
Diab & Sartawi [23]	Classification of learning outcomes into Bloom's taxonomy	Text similarity and NLP	Focused on learning outcomes rather than exam questions	No analysis of exam datasets

2. OBJECTIVE

2.1 Main Objective

To design, develop, and evaluate an AI-driven intelligent examination analytics framework that analyzes historical exam questions and student performance data using Natural Language Processing (NLP), topic modeling, and machine learning techniques to identify frequently misunderstood questions, detect conceptual learning gaps, and predict potential future exam topics with at least 80% prediction accuracy within a university Learning Management System environment.

- **Specific** – Develop an AI-driven exam analytics framework.
- **Measurable** – Target ≥80% prediction accuracy.
- **Achievable** – Uses existing technologies (NLP, ML, topic modelling).
- **Relevant** – Improves exam analytics and student learning strategies.
- **Time-bound** – Achieved within the research/project implementation period.

2.2 Specific Objectives

1. **To collect and preprocess historical examination questions and student performance data** from a university Learning Management System by applying data cleaning and text preprocessing techniques to prepare the dataset for analytical modelling.
2. **To apply Natural Language Processing (NLP) techniques to analyse exam question texts** and extract key concepts, keywords, and semantic relationships in order to identify the underlying topics present in examination papers.
3. **To implement topic modelling and clustering algorithms** to group similar exam questions and discover concept clusters that represent major subject areas and recurring themes in historical examinations.
4. **To integrate student performance data with analysed exam questions** in order to identify frequently misunderstood questions and detect conceptual learning gaps among students.
5. **To develop a predictive machine learning model** capable of analysing historical exam patterns and concept clusters to forecast potential future exam topics and question structures.
6. **To evaluate the performance and effectiveness of the proposed framework** by measuring prediction accuracy, clustering quality, and the usefulness of generated insights for improving teaching strategies and student exam preparation.

3. METHODOLOGY

3.1 Research Approach

This study adopts a Design Science Research (DSR) approach, which is suitable for developing innovative artifacts and evaluating them in real-world contexts. The framework will be designed, implemented, and iteratively refined to address the identified gaps in Predictive Learning Analytics (PLA) and Educational Data Mining (EDM). The DSR methodology ensures the development of a practical, usable, and scientifically rigorous AI-driven examination analytics system.

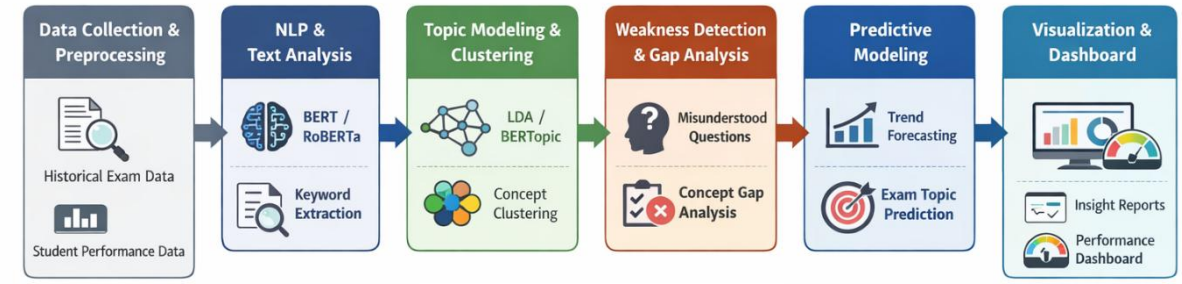


Figure 1 Process of the system

Technical Methods and Algorithms

1. **Data Preprocessing**
   - Historical exam papers and student performance records will be collected and cleaned.
   - Text preprocessing steps: tokenization, lemmatization, stopword removal, and normalization.
   - Structured student performance data (scores, attempts, timestamps) will be standardized and aligned with exam questions.
2. **Natural Language Processing (NLP)**
   - Extract semantic information from exam questions using transformer-based language models (e.g., **BERT or RoBERTa**).
   - Identify key concepts, question types, and cognitive skill levels (aligned with **Bloom's Taxonomy**).
3. **Topic Modelling and Clustering**
   - Apply **Latent Dirichlet Allocation (LDA)** or **BER Topic** for topic discovery from question texts.
   - Cluster similar questions based on semantic similarity using **K-Means** or **Hierarchical Clustering**.
   - Identify frequently recurring concepts and patterns in historical exams.
4. **Weakness and Gap Analysis**
   - Overlay student performance on clustered questions to detect frequently misunderstood questions.
   - Diagnose conceptual gaps and skill deficiencies using statistical analysis and cognitive mapping.

5. **Predictive Modelling**
    - Implement supervised machine learning models (e.g., **XGBoost, Random Forest, or Transformer-based predictive models**) to forecast likely future exam topics and question structures.
    - Incorporate historical trends, concept clusters, and topic recurrence rates into the predictive pipeline.
6. **Evaluation and Validation**
    - Evaluate predictive accuracy using **Precision, Recall, F1-Score, and overall prediction accuracy**.
    - Validate clustering quality using **Silhouette Score** and **Davies-Bouldin Index**.
    - Conduct user evaluation with instructors for usefulness and interpretability of generated insights.

3.2 Tools and Platforms

Table 3 Tools and platforms

Component	Tool / Platform	Justification
NLP / Question Analysis	Python, Hugging Face Transformers (BERT/RoBERTa)	State-of-the-art NLP models for semantic understanding and Bloom's level prediction.
Topic Modelling & Clustering	Python (BERTopic, Scikit-learn, NLTK, Gensim)	Efficient topic extraction, semantic clustering, and scalable analysis
Predictive Modelling	Python (XGBoost, PyTorch, TensorFlow)	Proven performance in supervised learning and sequence prediction tasks
Data Storage & Processing	PostgreSQL / MongoDB	Efficient storage of exam data, student responses, and performance metrics
Visualization & Dashboard	Plotly, Dash, or Streamlit	Interactive dashboards for instructors and students to interpret insights

Data Collection and Ethical Considerations

Data Sources: Historical exam questions, student scores, and LMS interaction data from the university's database.

Anonymization: All student data will be anonymized to protect privacy. No personally identifiable information will be stored in the system.

Ethical Compliance: Research will adhere to the university's ethical guidelines, and informed consent will be obtained where required.

Data Security: Access to raw data will be restricted to authorized researchers, and secure storage practices will be enforced.

Validation Strategy and Performance Metrics

Predictive Accuracy – Measure the ability of the framework to forecast likely exam topics using Accuracy, Precision, Recall, and F1-Score.
Clustering Quality – Evaluate concept clustering and topic discovery using Silhouette Score and Davies-Bouldin Index.
Weakness Detection Reliability – Compare the system's identification of misunderstood questions with instructor feedback and historical failure rates.
User Feedback – Conduct surveys with instructors and students to assess the usability, interpretability, and usefulness of the framework for targeted learning and curriculum planning.
Iterative Improvement – Refine models and algorithms based on validation results to maximize practical relevance and effectiveness.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

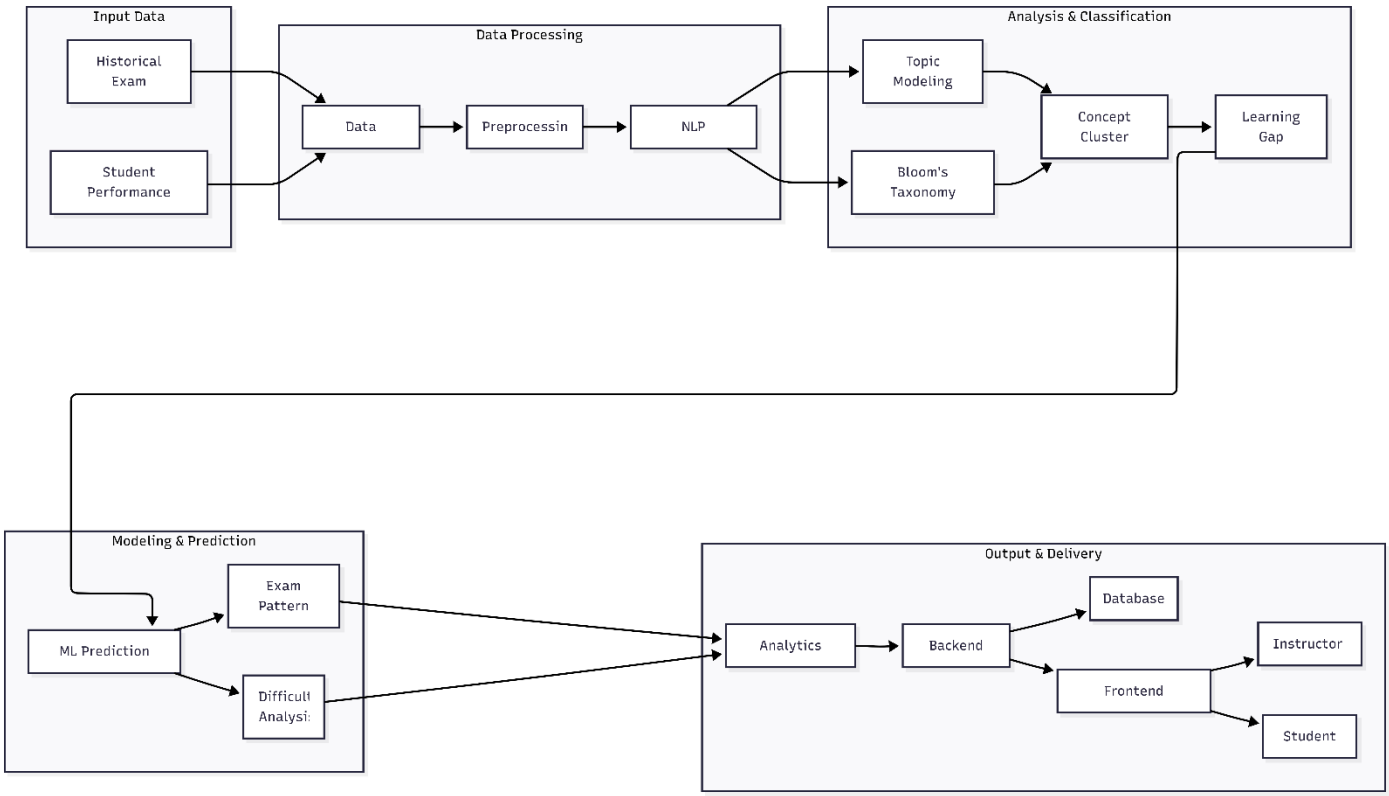


Figure 2 High Level Architecture

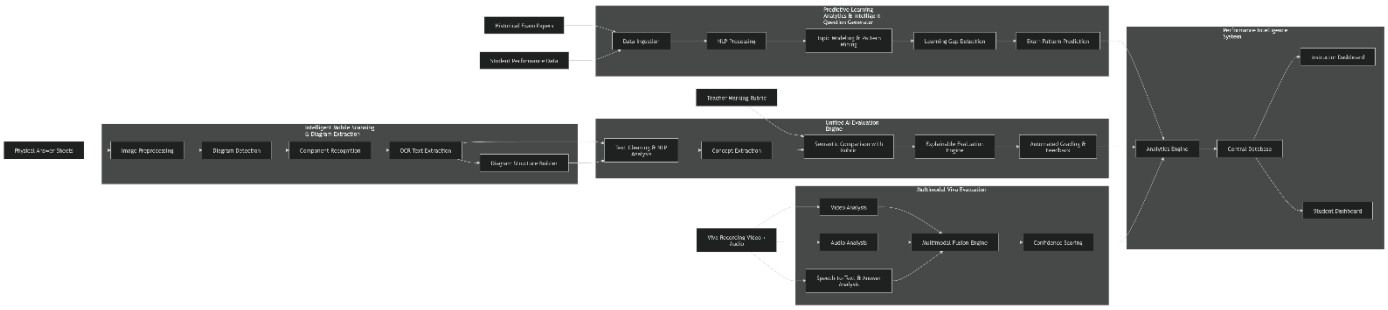


Figure 3 High Level Architecture All Function

5. USER REQUIREMENTS

Stakeholder Identification

The proposed Intelligent Examination Analytics System will involve several key stakeholders who interact with or benefit from the system.

Table 4 Stakeholder Table

Stakeholder	Role	Interest in the System
Students	Primary users	Identify weak concepts, understand commonly misunderstood questions, and prepare effectively for exams.
Lecturers / Instructors	System users	Analyze exam performance, identify conceptual gaps among students, and improve teaching strategies.
University Administration	Decision makers	Monitor overall academic performance and improve curriculum quality using data-driven insights.
System Administrators	Technical managers	Maintain the system, manage datasets, and ensure system reliability and security.
Researchers / Developers	System designers	Develop, maintain, and improve the AI models used for analytics and prediction.

Functional Requirements

The system must provide the following functional capabilities:

FR1: The system shall allow administrators to upload historical exam questions and student performance datasets.

FR2: The system shall preprocess uploaded exam data by cleaning, normalizing, and preparing textual content for analysis.

FR3: The system shall analyse exam questions using Natural Language Processing techniques to extract concepts and keywords.

FR4: The system shall apply topic modelling and clustering algorithms to group similar exam questions into concept clusters.

FR5: The system shall integrate student performance data with analysed exam questions to identify frequently misunderstood questions.

FR6: The system shall detect conceptual learning gaps based on student performance across different question clusters.

FR7: The system shall implement predictive analytics models to forecast potential future exam topics and question structures.

FR8: The system shall generate analytical reports and visual dashboards showing exam patterns, concept clusters, and performance insights.

FR9: The system shall allow lecturers to view insights related to student conceptual weaknesses and exam trends.

FR10: The system shall allow students to view personalized insights regarding their weak concepts and recommended study areas.

Non-Functional Requirements

Security Requirements

- The system shall ensure data privacy and confidentiality for all student information.
- The system shall implement role-based access control to restrict access to sensitive data.
- All stored data shall be securely encrypted and protected from unauthorized access.

Performance Requirements

- The system shall process and analyze uploaded exam datasets within acceptable processing times.
- The predictive models shall maintain high accuracy and reliability for exam topic predictions.
- The system shall support analysis of large historical exam datasets efficiently.

Usability Requirements

- The system interface shall be simple, intuitive, and user-friendly for both instructors and students.
- The system shall provide clear visualizations and dashboards to present analytical insights.

Reliability Requirements

- The system shall ensure consistent availability and reliable operation during data analysis tasks.
- The system shall maintain data integrity during storage and processing.

Scalability Requirements

- The system shall support increasing volumes of exam data and student performance records without significant degradation in performance

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona

The proposed Intelligent Examination Analytics System primarily targets higher education institutions, universities, online learning platforms, and educational training centres that conduct digital assessments and manage large volumes of exam data. These organizations often rely on basic grading systems that do not provide advanced analytics or predictive insights.

The key customer personas include:

University Lecturers / Instructors – Educators who design exams and evaluate student performance. They require insights into commonly misunderstood questions, concept difficulty levels, and student learning gaps to improve teaching strategies.

Academic Administrators – Faculty heads, department coordinators, and academic management teams who monitor academic performance and curriculum quality across courses.

Students – Learners preparing for examinations who need data-driven insights about frequently appearing exam topics and areas where they need improvement.

Educational Technology Providers – Companies developing Learning Management Systems (LMS) that can integrate advanced analytics modules to enhance their platforms.

Value Proposition

The proposed system offers unique value by converting traditional examination data into actionable academic insights. Unlike conventional LMS platforms that mainly display grades and performance summaries, the proposed framework provides intelligent analytics to improve both teaching and learning outcomes.

Key value propositions include:

- Identification of frequently misunderstood exam questions
- Detection of conceptual learning gaps among students
- Prediction of potential future exam topics based on historical patterns
- Data-driven teaching insights for instructors
- Personalized learning recommendations for students

These capabilities enable educational institutions to improve assessment quality and support more effective learning strategies.

Revenue Model

The system can be commercialized using a **Software-as-a-Service (SaaS)** business model where educational institutions subscribe to the platform.

The primary revenue streams include:

- **Institutional subscription plans** for universities and colleges
- **Integration services** for existing Learning Management Systems
- **Premium analytics features** offering advanced predictive insights
- **Customization and consulting services** for institutions requiring tailored analytics solutions

This model allows the platform to scale efficiently while providing continuous updates and improvements.

Cost Estimation

The development and deployment of the system involve several cost components:

Table 5 Cost Estimation

Cost Category	Description
Development Costs	Software development, AI model development, and testing
Infrastructure Costs	Cloud hosting, storage, and computing resources
Data Processing Costs	Processing large datasets for machine learning training
Maintenance Costs	System updates, technical support, and bug fixes
Marketing and Deployment	Promotion, onboarding institutions, and user training

Initially, the system can be developed using open-source technologies such as Python, machine learning frameworks, and cloud services to reduce development costs.

Pricing Strategy

The pricing strategy will be based on a tiered subscription model, allowing institutions of different sizes to access the system.

Example pricing tiers:

- **Basic Plan** – Basic exam analytics and performance dashboards for small institutions.
- **Professional Plan** – Advanced analytics including concept clustering and predictive exam analysis.
- **Enterprise Plan** – Full system features including API integrations, custom analytics modules, and dedicated support.

Pricing may vary depending on number of users, dataset size, and institutional requirements.

Competitive Advantage

The proposed system provides several advantages compared to existing educational analytics platforms:

- **Question-level analytics** rather than only overall student performance analysis.
- **Integration of NLP and machine learning** to analyse exam question structures.
- **Predictive exam pattern analysis**, which is rarely available in existing educational analytics systems.
- **Concept clustering and gap detection**, allowing educators to identify conceptual weaknesses more effectively.
- **Comprehensive analytics framework** that combines exam content analysis with student performance data.

These features provide a unique advantage compared to traditional learning analytics tools that focus mainly on performance dashboards.

Intellectual Property Considerations

The intellectual property of the proposed system will primarily focus on the AI models, algorithms, and system architecture developed for exam analytics and predictive assessment.

Possible intellectual property protections include:

- **Copyright protection** for the developed software platform and source code.
- **Patent opportunities** for novel algorithms used in predictive examination analytics and question clustering.

- **Trade secrets** related to proprietary data processing pipelines and machine learning models.

If commercialized, the system may also include **licensing agreements** with educational institutions or LMS providers to ensure controlled use and distribution of the technology.

7. BUDGET AND JUSTIFICATION

Table 6 Budget and Justification

Category	Item	Estimated Cost (LKR)	Justification
Software Licenses	Data Analysis & Development Tools	Rs. 0	Tools such as Python, TensorFlow, PyTorch, Scikit-learn.
	Hugging Face Subscription (10 months)	Rs.30,000	The premium AI model that fast and reliable for train large database
Cloud Services	Cloud Computing (Model Training & Deployment)	Rs. 85,000	Cloud platforms provide scalable computing resources for training machine learning models.
	Cloud Storage	Rs.30,000	Storage for datasets, trained models, and application data.
Miscellaneous	Internet Connection (8 months)	Rs.10,000	Stable internet connection for train and develop the environment
	Data Collection, and Testing	Rs.10,000	Supports online research, data acquisition, and testing activities.
Total		Rs.165,000	

8. WORK BREAKDOWN STRUCTURE (WBS)

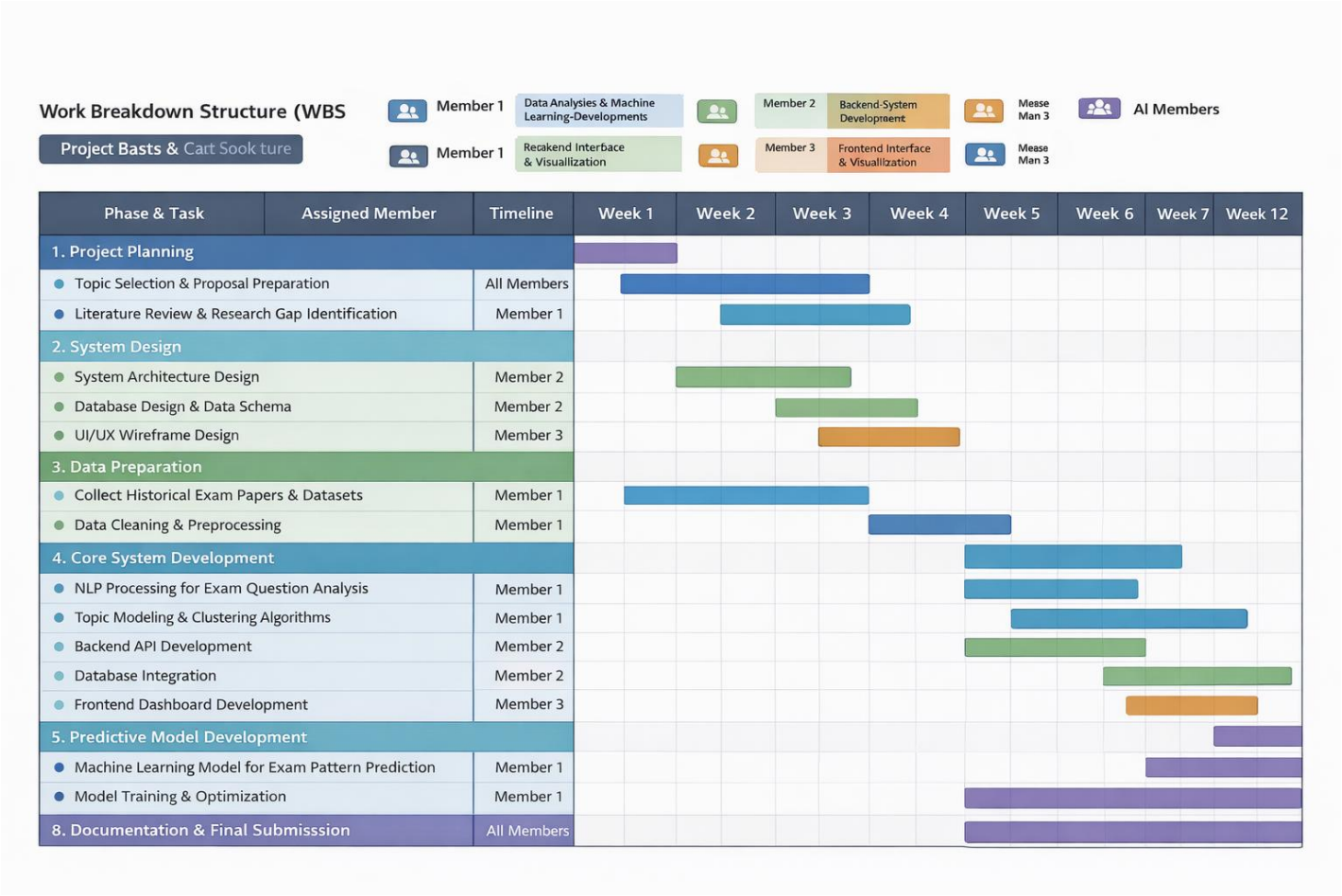


Figure 4 Work Breakdown Structure (WBS)

9. GANTT CHART



Figure 5 Gantt Chart

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